

Engineered Observation Limits in Exact Finite Models

What an Information-Theoretic Audit Certifies, Where It Stops, and Where It Does Not

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Cryptography is the engineering discipline of declared observation limits. Where the rest of this campaign has to find the consumer's channel inside a physical model, cryptography writes the channel down on purpose and then argues about what a consumer can read through it. This paper carries the campaign's consumer-relative audit to that argument in eleven registered runs, each on a declared toy model small enough to enumerate, and to nothing else. Nothing in this paper bears on computational hardness or on asymptotic security, no real cipher, protocol, curve, standard, implementation, device, or deployed system is modeled or evaluated, and no attack is developed. What is measured is evaluation methodology, meaning the reasoning that carries a leakage measurement to a conclusion about what a reader can do. A threshold secret-sharing scheme over a field of thirty-one elements, enumerated by integer counting over all 29791 outcomes, gives an information curve that is a step, exactly zero below the declared threshold with the reader's posterior exactly the prior and exactly the whole secret at and above it, with no subset of any size in between. The redundancy plateau the campaign found in an environment rather than placing it takes a different value at every one of its six fragment sizes, from 0.8118 to 1.9996 bits, so the two differ in shape and the difference is that one of them was designed. A declared control with short randomness carries 2.6323 bits to each of its five single-share readers, so the zeros belong to the scheme rather than to the measurement. Declared leakage channels are experiments in Blackwell's sense, and whenever two of them read the same hidden quantity a strictly positive conditional information proves non-garbling with no tolerance and no search. The channel that reads a Hamming weight carries more than twice the information of the channel that reads one bit and is still not above it in Blackwell's sense, the two conditional informations measuring 1.8113 and 0.7806 bits. In two thousand declared random channel pairs, 268 were ordered unanimously by all seven declared divergences while proved incomparable, and every one of them carried a declared task that strictly prefers the channel every divergence ranks as the less leaky, with no such reversal among the 867 comparable pairs. The measured statement is that a reduction in a scalar leakage measure is not a guarantee for a reader's task outside certified comparability, and its constructive half is that the same certificate decides comparability by proof, so an evaluation that establishes comparability may rely on its scalar ordering. Two consumers price one declared source at 3.477 bits and at exactly 1 bit, both correct, the disagreement lying entirely in the task. One of those runs failed, on a declared clause requiring the difference of the two accountings to increase along a ladder of sources. The difference turned over at the far end, because both accountings vanish together as a source becomes certain, and the corrected registration measures the ratio, which falls strictly where the difference does not. The arc then measures where an audit of this kind stops. A declared generator over a toy curve was given two parameter sets, one whose relating scalar is known by construction and one whose scalar is fixed by a declared rule and recovered afterwards by enumeration. That run failed two bars, because a group of ninety-six elements cannot spread ninety-five seeds near uniformly over eight output words, measuring 2.7890 and 2.8002 bits against three, and because the recovery loop accepted the first candidate consistent with one observed word instead of testing it against the next, predicting at 0.263 against a declared half. It also asked the wrong question, since every point of a cyclic group has such a scalar and the run recovered the second one by enumeration. A second registration on a larger group of prime order failed one bar from a coding error in an inverse map, measuring 0.0314 where the bar was one part in a billion, with its positive control already exact. The corrected registration passes. A declared one-bit covert channel carries exactly 1 bit of information about the seed bit it was given while the unmodified generator carries exactly 0, both parameter sets produce output words at 2.9878 and 2.9943 bits against a uniform three, and the two routes to one conditional entropy agree exactly. A trapdoor scalar is a constant of the parameter set, so its entropy is exactly zero and the information it can carry about any observable is bounded by that zero. That is a bound on what such an audit can see and not advice about anything. What such an audit sees instead is counted next. Over all 10100 non-singular curves of a toy parameter space, 2275 lie in a declared rare class, a density of 0.2252. A procedure allowed one seed reaches one curve and did not land in the class. A procedure allowed 4096 seeds reaches only 99 distinct curves, so its nominal freedom overstates the freedom it has, and it reaches the class at its tenth seed with 526 of its seeds landing there. The published artifact is a curve and a seed and carries no record of how many seeds were tried, so only a procedure's declared freedom is auditable, which is preregistration under another name. The last two runs measure a place where none of this stops. A consumer reading four declared preparations, two of which are not

orthogonal to the other two, gains information only by disturbing what it reads, running from 0.05 bits at an induced error of 0.025 to 0.5 bits at 0.25, agreeing with declared closed forms to sixteen decimal places, at an exchange rate of exactly two bits of information for each unit of induced error. The same structure on an orthogonal ensemble reads the whole bit at an induced error of exactly zero, which is what the campaign's hydrodynamic track measured when a coarse observer's decayed mode was recovered bit for bit. A declared classical record broadcast to two consumers gives each of them the whole bit, so their informations sum to two, while a declared three-qubit family gives them correlations whose squared strengths sum to one across the whole grid to ten decimal places, so a consumer holding all of it leaves the other none. Redundancy is available exactly for information that has already become classical.

I. INTRODUCTION

This campaign models observation as consumption. A consumer reads a hidden state through a declared channel, spends a declared budget on the read, and acts on a declared task, and every claim is settled by exact measurement in a declared finite model against bars fixed before the run [14, 15]. Cryptography is the one engineering discipline that already works this way. Elsewhere the campaign has to find the consumer's channel inside a physical model and argue that the choice was not made to suit the answer. Cryptography writes the channel down on purpose, in a specification, before anyone measures anything.

Three of its central objects are campaign objects under other names. Semantic security is restricted distinguishability, meaning a statement that a reader with declared powers cannot tell two things apart [4]. A side-channel adversary is a consumer with a declared channel, which is what the campaign calls a restricted observer [7]. Threshold secret sharing is an engineered redundancy plateau, built by hand where the campaign's quantum track found one in an environment [2, 15]. Shannon's information-theoretic account of secrecy is the oldest statement in the campaign's own language, a statement about what a reader can recover from what a reader can see [1].

The limit of what can be said here is stated once in the abstract, once here, and once in the ledger of Sec. X. This paper measures exact finite models, so it can say nothing about computational hardness, nothing about asymptotic security, and nothing about any concrete algorithm or deployment. Every model is a declared toy model. No real cipher, protocol, curve, standard, implementation, device, or deployed system is modeled or evaluated. No attack is developed and no attack is evaluated. What remains is evaluation methodology, meaning the reasoning that carries a leakage measurement to a conclusion about what a reader can do, and that reasoning can be audited exactly in settings where every alternative reading is computable rather than arguable [8].

Three claim types run through the paper and are never conflated. Instrument claims state that leakage informativeness, garbling relations between declared channels,

and optimal reads can be computed exactly in finite models, and they are checked against closed forms. Structural claims state that a named evaluation statement is channel-relative or measurably is not, inside a declared model with every alternative channel enumerated, and they carry the campaign's exploratory label. Claims about real systems are absent. No run in this paper is labeled as supporting one, and the limit of the previous paragraph is the reason. A reader looking for evidence about the strength of a deployed system will find none here.

An anti-circularity contract governs every run. No leakage model was chosen after inspecting which choice favors a wanted verdict. No adversary task was selected after seeing the answer. No divergence or entropy measure was added to a battery after seeing the battery's verdict. Channels, tasks, budgets, priors, ladders, seeds, and bars were written into the track document before the runs they govern. One bar was moved before its run, because the object it was drafted against provably could not meet it, and the move and its reasoning were recorded in the track document before the run rather than afterward. Sec. II reports what that object measured.

The paper reports eleven registrations in three movements, and three of the eleven are corrections of an earlier registration that failed its own declared bar.

The first movement measures what an audit of this kind can certify. It runs the instrument layer on an engineered redundancy plateau computed by exact integer counting and validated against the known answer for threshold secret sharing, orders declared leakage channels by Blackwell's relation and replaces a search with a proof, measures where a scalar leakage ordering and a task ordering disagree, and prices one declared source for two consumers with different tasks.

The second movement measures where the same audit stops. A trapdoor in a declared generator is a constant of the parameter set rather than a source of randomness, so its entropy is zero and no quantity built from informations can see it, which is a bound rather than a shortcoming of any instrument. What the audit can see instead is provenance, and provenance turns out to be a counting question about how many results a procedure was free to choose among.

The third movement measures a place where none of that stopping happens. When the states a consumer reads are not distinguishable, a consumer that learns anything changes what it read, and the price is exact. When two

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consumers want the same correlation, one of them having all of it leaves the other with none, while a classical record is held in full by both at once. These are the two runs where the campaign's own instruments have no ceiling, and they are what the earlier movements exist to locate.

II. AN ENGINEERED REDUNDANCY PLATEAU IS A STEP AND ONE LEFT BY AN INTERACTION IS NOT

The substrate is Shamir's threshold scheme over the prime field with 31 elements, with threshold 3 and 5 shares [2]. A secret uniform on the field and two uniform independent coefficients determine a share vector at the evaluation points 1 through 5, so every distribution in this section is a function of $31 \times 961 = 29791$ equally weighted outcomes. Every distribution was computed by enumerating all of them with integer counting and never by sampling. Informations are built from integer count ratios, so a ratio that is exactly one contributes exactly zero, and every summation over outcomes uses compensated addition. The declared tolerance is 10^{-12} wherever a tolerance is needed at all.

The measured information curve as a function of the number of shares read is

$$(0, 0, \log_2 31, \log_2 31, \log_2 31) \text{ bits}, \quad (1)$$

with $\log_2 31 = 4.954196310386875$. All 15 subsets below the declared threshold carry mutual information exactly 0.0 bits rather than merely inside the tolerance, and for every share value of positive probability the reader's posterior on the secret is the uniform prior with deviation exactly 0.0 at every field element. All 10 threshold subsets carry exactly $\log_2 31$ bits with conditional entropy exactly 0.0, and an independent route, Lagrange interpolation at zero over the field, recovered the secret in all 297910 subset-outcome checks with zero failures. The exactness is structural rather than fortunate. Below the threshold every count ratio in the sum is the integer one, and at the threshold every posterior is a point mass. The step is a step in the strongest available sense, the maximum distance from any of the 31 nonempty subsets to the nearer of the two plateau values being exactly 0.0, so no subset of any size takes an intermediate value. Mutual information computed from the joint distribution and mutual information computed as a difference of entropies agree across every subset tested in all three schemes of this section to 1.8×10^{-15} .

Figure 1 places that curve beside the plateau the campaign found rather than placed. In the quantum track a system interacts with an environment and an observer reads a fragment of the environment, and the information a fragment carries rose through 0.8118, 0.9571, 0.9998, 1.0426, 1.1878, and 1.9996 bits as the fragment grew from one environment qubit to six, a different value at every size [15]. The height of that plateau and the speed of the approach to it are set by how strong the records written

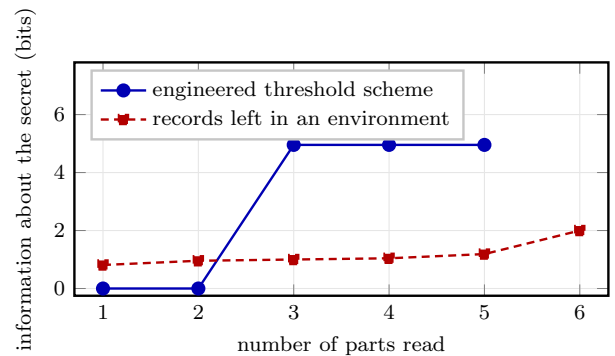


FIG. 1. Two redundancy plateaus. The engineered curve is the mutual information between the secret and a subset of shares in the declared threshold scheme, exactly zero at one and two shares and exactly $\log_2 31$ bits at three, four, and five, with no subset of any size taking an intermediate value. The other curve is the campaign's earlier measurement of how much a fragment of an environment carries about a system after a declared interaction has written records into it, rising from 0.8118 to 1.9996 bits with the number of environment qubits read and taking a different value at every fragment size. Both curves are read from committed evidence records.

by the interaction are. The engineered plateau has no approach at all. It is a step, and it stands exactly where the designer put the threshold. The comparison is between two declared models and is not a statement about physical systems.

The declared negative control decides whether the zeros belong to the scheme or to the measurement. The control is a short-randomness scheme, a degree-one polynomial whose single coefficient is uniform on the declared five-element set rather than on the field, with shares at the same five evaluation points. Its declared bar required at least one single-share reader to carry more than 0.1 bits. Every one of its five single-share readers carries

$$\log_2 \frac{31}{5} = 2.6322682154995127 \text{ bits}, \quad (2)$$

matching the closed form with deviation exactly 0.0 and clearing the declared bar by more than a factor of 26, and its two-share readers all sit at $\log_2 31$, the same step one rung earlier. The instrument sees leakage when leakage exists.

The drafted control is the recorded deviation and the section's second finding. It was a degree-one polynomial with a field-uniform coefficient whose five shares are published together with the redundant value formed by adding the secret to the coefficient. That scheme cannot leak to any single reader, because every one of its six published values is the secret masked by an independent field-uniform quantity and is therefore exactly uniform and exactly independent of the secret, and its sixth value is bit for bit the first share. A bar whose outcome is settled before the run tests nothing, and this one was settled against the drafted scheme, so the bar moved to the

TABLE I. The five declared leakage channels, what each carries about the declared key, and the certificate that decides its position under the identity channel. The conditional information of key and channel given the identity reading is exactly zero for all four non-identity channels, which proves each is a garbling of the identity. All values are from the committed evidence record.

Channel	information (bits)	conditional information given the identity (bits)
identity	4	
Hamming weight	2.0306	0
low bit	1	0
noisy Hamming wt.	0.8867	0
constant	0	0

short-randomness scheme before the run, and the drafted scheme was measured without a bar with its outcome predicted in advance so that the run could refute the prediction. It did not. All six single-share informations came out exactly 0.0, the sixth published value was identical to the first share at every one of its 961 outcomes, and exactly one of the 15 pairs of published values is uninformative, the pair of identical values, with the other 14 determining the secret. Publishing a redundant copy of a share leaks nothing to any single reader. Redundancy and leakage are different things, which the campaign’s quantum track reached from the other side when it found that redundant records are what make a state readable by many observers at once [15].

III. A CONDITIONAL INFORMATION CERTIFICATE PROVES NON-GARBLING WITHOUT A THRESHOLD

A declared key-dependent leakage model is an experiment in Blackwell’s sense, a row-stochastic matrix whose rows are the reading distributions in each hidden state [3]. Blackwell’s theorem says that one experiment is at least as good as another for every prior and every task exactly when the second is a garbling of the first, meaning the second can be produced from the first by post-processing that does not see the hidden state. The question of whether one declared side channel is better than another therefore has a precise answer with two possible forms. Either one garbles the other, and then the ordering holds for every task a reader might have, or neither garbles the other, and then no single number can order them for every task.

The substrate is a four-bit key uniform on sixteen values, a declared bijective substitution table, and five declared channels that read the substituted value. The identity channel reads the value. The Hamming-weight channel reads the number of set bits. The low-bit channel reads the lowest bit. The noisy Hamming-weight channel adds declared symmetric noise of probability 0.2 in each direction to the Hamming weight. The constant channel reads nothing. Table I gives what each carries about the key.

The certificate is the transferable part of this run, and

it replaces a search with a proof. Channel B is a garbling of channel A exactly when B is conditionally independent of the key K given A , so

$$I(K; B | A) = 0 \iff B \text{ is a garbling of } A, \quad (3)$$

and a strictly positive left-hand side proves that B is not a garbling of A with no tolerance and no search involved. The certificate needs a joint distribution of the key with both channels, which exists here because every declared channel reads the same substituted value. Two channels given only as separate matrices have no such joint, and for those the campaign’s earlier tools apply instead, a conic lower bound on the garbling residual and the monotonicity of an f -divergence. The projected-gradient garbling search used elsewhere in the campaign was retained here only as a cross-check. An earlier run in the campaign established incomparability by thresholding that search residual, the threshold was chosen without computing the scale it would be compared against, the run failed its own bar, and a separate registration had to repair it with an analytic bound [14]. Equation (3) removes the threshold from the argument entirely.

All five declared items passed. The identity channel is above every declared channel, each search residual exactly zero and each conditional information given the identity exactly zero. The constant channel is below every declared channel, each residual exactly zero as well. Adding the declared noise to the Hamming weight is a garbling, residual 2.2×10^{-16} and conditional information 4.8×10^{-17} , and the reverse direction is proved impossible without any tolerance argument, the conditional information of key and Hamming weight given the noisy reading measuring 1.1439 bits, which is exactly the difference between what the two channels carry. The data-processing control holds across the declared battery of 201 key-recovery tasks, 200 seeded random utilities on sixteen actions plus the maximum-likelihood key-guessing task, with a worst task gain under garbling of 5.6×10^{-17} .

The remaining pair is the incomparable one. The Hamming-weight channel carries 2.0306 bits about the key and the low-bit channel carries exactly 1, so a scalar comparison ranks the first above the second by better than a factor of two. Yet 1.8113 bits of the Hamming weight survive knowing the low bit and 0.7806 bits of the low bit survive knowing the Hamming weight, so neither is a garbling of the other and no scalar comparison orders them for every task. Carrying more than twice as much information and being better for every task are different properties, and in this declared model the first holds while the second fails. The next section measures how often the two part company and what happens to a reader’s task when they do.

IV. UNANIMOUS SCALAR LEAKAGE ORDERINGS CARRY NO TASK GUARANTEE OUTSIDE CERTIFIED COMPARABILITY

A single declared secret bit, uniform, is read through channels into a declared four-symbol alphabet, so every channel is a two by four row-stochastic matrix. The declared scalar battery compares the two conditional reading distributions with seven f -divergences, the two Kullback-Leibler directions, total variation, squared Hellinger, the two chi-squared directions, and Jensen-Shannon [6]. The declared task battery holds 205, two hundred seeded random utilities on three actions and five commitment tasks in which a reader may either decline or guess the secret bit, gaining 1 for a correct guess and paying a declared cost of 2, 5, 10, 20, or 50 for a wrong one. A task reversal is a pair of channels in which all seven divergences strictly order one channel above the other and some declared task strictly prefers the other by more than the declared 10^{-6} .

The declared exhibit contrasts two channels that put their separation in different places. Channel A has rows (0.55, 0.25, 0.15, 0.05) and (0.45, 0.25, 0.15, 0.15), so its two reading distributions differ by 0.10 on the most likely symbol and by 0.10 on the least likely one and agree exactly on the other two. Channel B has rows (0.49, 0.49, 0.01, 0.01) and (0.49, 0.49, 0.005, 0.015), so its two reading distributions agree exactly on the two symbols carrying almost all the mass and differ by 0.005 on each of the two rare symbols, which are equally likely under one value of the secret and stand in the ratio one to three under the other. The exhibit was declared before the run so that the run could refute it. All seven divergences order A above B , the smallest margin being 0.0149, and a declared task prefers B by 0.00212, so the exhibit met its bar. The pair is proved incomparable in both directions, the conic lower bound on any garbling residual measuring 0.00175 in one direction and the total-variation difference measuring 0.095 in the other, a garbling being unable to raise an f -divergence. Neither of those two numbers is a search residual.

Two further items were declared with the exhibit. The data-processing control drew five hundred declared garbled pairs and checked them against the full task battery. No divergence rose under any garble, the smallest drop in the required direction measuring 0.00262, and no task value rose by more than 3.3×10^{-16} , which is Blackwell's theorem and the data-processing inequality holding inside the instrument. The localization item classified a declared ensemble of 2000 seeded random channel pairs by comparability and required every task reversal to fall in the incomparable class. Figure 2 shows the outcome, 867 comparable, 856 incomparable, 277 ambiguous, and zero task reversals among the comparable pairs, with 210 among the excluded ambiguous pairs. The prevalence measurement was declared with no bar. Among the 856 certified-incomparable pairs, 268 are ordered unanimously by all seven declared divergences, and every one of those 268 carries a task reversal in the declared battery,

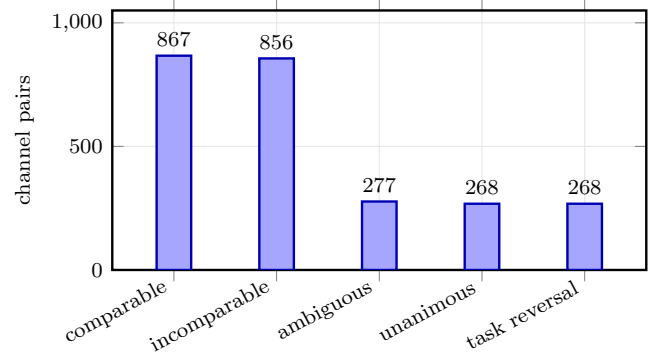


FIG. 2. The declared ensemble of 2000 random leakage channel pairs. The first three bars partition the ensemble by comparability, comparable when one of the two garbling searches lands inside the instrument's feasibility tolerance, incomparable when both leave a residual above the declared 0.01, ambiguous otherwise and excluded as declared. The last two bars are subsets of the incomparable class, pairs ordered unanimously by all seven declared divergences and pairs whose unanimous ordering is reversed by a task in the declared battery. The two subsets are equal at 268. No comparable pair carries a reversal. Every count is read from the committed evidence record.

the largest reversal margin being 0.0796 and the median 0.0183. The campaign's decision-theoretic track measured the same structure in a different substrate and found 281 of 281 [14].

The reading was fixed in the track document before the run and is binding on this report. A reduction in a scalar leakage measure is not a guarantee for a reader's task outside certified comparability, because in every unanimously ordered incomparable pair measured here the ordering the scalar measures report and the ordering a task reports disagree. This is a statement about evaluation methodology in declared finite models. It is not a claim that any countermeasure, implementation, or system is insecure, and none was modeled. The constructive half of the statement matters as much and is measured by the same runs. Comparability is common rather than rare in this ensemble at 867 of 2000 pairs, and not one comparable pair carried a reversal, so an evaluation that first establishes comparability may rely on its scalar ordering for every task. Establishing it is cheapest when the two channels read the same hidden quantity, because Eq. (3) then settles the question by proof. The channels of this ensemble are drawn as separate matrices with no shared reading, so they were classified by the garbling search against the declared tolerances instead.

V. THE DIFFERENCE BETWEEN TWO ENTROPY ACCOUNTINGS TURNS OVER AND THEIR RATIO DOES NOT

The same declared source is read by two consumers. One averages over many independent reads and is charged the Shannon entropy. One makes a single guess and is charged the min-entropy. Both accountings are functions of the source alone, and which of them prices the source correctly is set by the consumer's task. This run measures where the two part company, and it is the run that failed a declared bar.

The substrate is two declared families of sources on an alphabet of 32 outcomes, every probability an exact rational. Family U places $1/32 + \epsilon$ on outcome zero and spreads the rest uniformly, with ϵ on the declared ladder 0, 0.01, 0.05, 0.15, 0.30, 0.60, 0.90, so the first member is exactly uniform and the last is a hard spike. Family S places q on outcome zero and spreads the rest uniformly, with q on the declared ladder 0.5, 0.9, 0.99. The declared key-material case is the member of S at $q = 0.5$, half the mass on one outcome and half spread over the remaining 31. Every distribution is built from exact rationals so the probabilities sum to exactly one and the code construction compares rationals rather than rounded floats.

Three instrument items passed and were unchanged by the correction. Summed Shannon entropy agrees with its closed form at every one of the ten declared sources to 8.9×10^{-16} , and the min-entropy computed from the exact numerator and denominator of the largest probability agrees with the direct route to the same 8.9×10^{-16} , with the largest probability located by exhaustive scan over the alphabet rather than assumed. The optimal single-guess success probability, found by exhaustive maximization over the alphabet, equals two to the power minus the min-entropy with worst deviation 1.4×10^{-17} , and eight of the ten sources are bit for bit identical. A Huffman code was constructed for every source with a heap and no library import [5], every code satisfies the Kraft equality exactly, and every mean code length sits at or above the Shannon entropy and strictly below the entropy plus one, the classic bound [6]. The smallest lower margin is exactly 0.0 at the uniform member, where the mean length is exactly 5 bits and the entropy is exactly 5 bits, and the smallest upper margin is 0.0807 bits at $q = 0.99$, where the mean length 1.0497 sits below the entropy plus one of 1.1303.

The failed item is the fourth, and the failure is in one clause of the declaration rather than in the substrate. The clause declared that the min-entropy is at most the Shannon entropy at every source, that the two agree exactly at the uniform member and nowhere else, and that the difference between them increases strictly along the ladder. The first two held, the worst value of min-entropy minus Shannon entropy measuring exactly 0.0 and the equality holding exactly at the uniform member. The third failed. The measured difference along the ladder

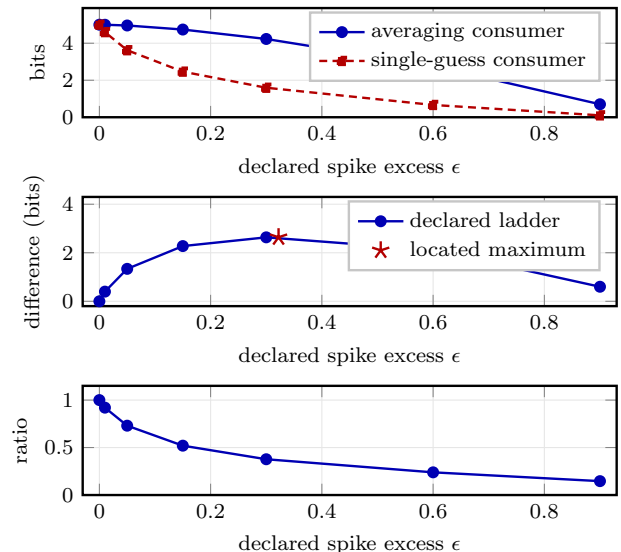


FIG. 3. Two accountings of one declared source family along the declared ladder. The top panel shows what the averaging consumer and the single-guess consumer are charged. The middle panel shows the number of bits separating them, which rises for four ladder steps and falls for two, with the star marking the interior maximum located by bisection on the closed-form derivative at spike weight 0.35338, meaning a spike excess of 0.32213, where the separation reaches 2.6398 bits. The declared ladder came within 0.0045 bits of that maximum and stepped over it. The bottom panel shows the fraction of the averaging consumer's accounting that survives to the single-guess consumer, which falls strictly at every ladder step. All values are read from committed evidence records.

is

$$0, 0.398, 1.337, 2.275, 2.635, 2.113, 0.599 \text{ bits}, \quad (4)$$

so the first four increments are positive and the last two are negative, and the recorded verdict of the run is FAIL.

The design error is named in the record. The declaration took a fact about the near-uniform end of the ladder and extended it to the whole ladder without checking the other end. Both accountings go to zero together as the spike takes all the mass, because a source that is almost certain is cheap to a consumer who averages and cheap to a consumer who guesses once, so the difference has to turn around and its maximum is interior. The declared ladder crossed that maximum between $\epsilon = 0.30$ and $\epsilon = 0.60$. The difference of the two accountings was the wrong quantity to demand monotonicity from.

The correction replaced the refuted clause with a monotone statement the substrate supports and added the location of the interior maximum as a measurement, changing nothing else. The ratio of the single-guess accounting to the averaging accounting falls strictly along the whole ϵ ladder, through 1.0, 0.9203, 0.7304, 0.5199, 0.3769, 0.2390, 0.1464, and strictly along the whole q ladder, through 0.2876, 0.1576, 0.1112. The closed-form derivative of the

difference with respect to the spike weight a is

$$\frac{d}{da} [H(a) - H_\infty(a)] = \log_2 \frac{1-a}{31a} + \frac{1}{a \ln 2}, \quad (5)$$

which reads 46.166, 34.559, 16.301, 5.181, 0.4147, -3.444 , and -7.165 along the ladder, one sign change, falling inside the bracket declared for it before the run. That derivative was checked against a central finite difference of the measured difference at every ladder point and agreed to 1.6×10^{-8} against the declared 10^{-6} , so the unimodality statement rests on a derivative that was checked against the quantity it differentiates. Bisection inside the declared bracket converged in 39 steps to a width of 5.5×10^{-13} and places the maximizing spike weight at 0.35338, where the separation reaches 2.6398 bits and the derivative reads 4.5×10^{-12} . The largest separation on the declared ladder was 2.6353 bits, so the ladder came within 0.0045 bits of the true maximum and stepped over it. The corrected registration passed all three of its items. Figure 3 shows all of it.

The measurement that carried no bar in either registration is the key-material case. For a source with half its mass on one outcome and half spread uniformly over the remaining 31, the averaging consumer is charged 3.477098155193437 bits and the single-guess consumer is charged exactly 1 bit, a factor of 3.477 between the two prices. The single guess succeeds with probability exactly 0.5 and a Huffman code for the same source averages 3.4839 bits per symbol. Both numbers are correct readings of one substrate and the disagreement is entirely in the task. Which entropy is the right entropy is set by the consumer's task and not by the source. Nothing in this section is a claim about any real key-generation system, and no such system is modeled.

VI. A CONSTANT HAS ZERO ENTROPY AND AN AUDIT BUILT FROM INFORMATIONS IS BLIND TO IT BY THAT BOUND

Everything so far measured what an audit built from informations certifies. The next two sections measure where it stops, and the first of the two stops it with a bound rather than with a failed attempt.

The object here is an asymmetry between two readers of one output stream who see the same words and one of whom holds a scalar the other does not. The substrate is a declared generator over a toy curve, chosen small enough that every point and every state can be enumerated. The state advances by one point multiplication and the output word is a declared truncation of the abscissa of a second multiplication, keeping three bits, so there are eight possible output words. Two parameter sets were declared, one whose second point is a known multiple of the generator and one whose second point is fixed by a declared rule and whose relating scalar is recovered afterwards by enumeration only so that the run can state it.

TABLE II. The three registrations of the backdoor-ceiling question. The first two failed and are kept. The output entropies are measured against a uniform three bits, the declared covert channel writes one seed bit into the output word, and the two routes are the public route to a conditional entropy and the route available to a reader holding the scalar. Verdicts and values are read from the committed evidence records.

	CR-4	CR-4b	CR-4c
verdict	FAIL	FAIL	PASS
group order	96	1013	1013
output entropy, known scalar	2.7890	2.9878	2.9878
output entropy, declared rule	2.8002	2.9943	2.9943
declared channel (bits)		1	1
unmodified generator (bits)		0	0
route difference (bits)		0.0314	0
scalar entropy (bits)			0

The first registration failed two of its four bars, and Table II carries all three registrations together. Its group has 96 elements and its declared truncation has 8 possible output words, so 95 seeds spread over 8 words cannot come near uniform, and the measured output entropies of 2.7890 and 2.8002 bits missed the declared window of 0.15 bits below a uniform 3 by finite size alone. The reader holding the scalar predicted the next word from one observed word at a rate of 0.263 against a declared half, because the recovery loop accepted the first candidate consistent with one observed word instead of testing candidates against the next word. Both are errors of declaration and implementation rather than properties of the substrate, and both are recorded.

The same run refuted its own question. The declared finding stated that the audit cannot separate a parameter set whose scalar is known from one whose scalar is merely unknown, and it read False. Two different second points induce genuinely different exact output distributions, and the two recorded word-count multisets are not the same multiset, so distributional identity was never what the two parameter sets share. What they share is that the scalar of the second point was recovered by enumeration in the same run, at the value 41. Every point of a cyclic group is a multiple of the generator, so the two parameter sets are not two classes of object at all, and there is nothing for a separation test to separate.

The corrected question replaces separation with blindness. A trapdoor scalar is a fixed constant of the parameter set rather than a source of randomness, so it cannot change any conditional law of the outputs and its whole content is the cost of a computation. An audit built from informations should therefore be exactly blind to it, and the measurement is that blindness placed beside a positive control that the same audit does detect. The second registration moved to a declared curve over the field of 1009 elements whose group has prime order 1013, which makes near uniformity attainable, and its positive control came out exact. A declared generator that writes one seed bit into its output word carries exactly 1 bit of mutual information with that seed bit, and the unmodi-

fied generator carries exactly 0. The output entropies are 2.9878 and 2.9943 bits against a uniform 3, and the scalar of the second point was again recovered by enumeration, at the value 2.

That registration failed on one bar and the cause is a coding error that the record names. Two routes to one conditional entropy measured 2.8811 and 2.9125 bits, a difference of 0.0314 where the bar was 10^{-9} . The route available to a reader holding the scalar multiplied the observed point's abscissa as though the abscissa were itself a scalar, where the group law requires scalar-multiplying the observed point by the inverse of the trapdoor scalar. The two routes are equal by the group law, so the measured disagreement was a measurement of the coding error and of nothing about trapdoors.

The third registration corrects the inverse map and passes all five of its items. The two routes now agree exactly, difference 0.0 rather than a small number, on the same substrate and with the same output entropies and the same positive control. The ceiling itself is restated in the form that makes it a bound. Write T for the trapdoor scalar and X for any observable an audit can form from the output stream. Then

$$I(T; X) \leq H(T) = 0, \quad (6)$$

because a constant of the parameter set has exactly zero entropy, and the record carries both quantities at exactly zero. The blindness of an audit built from informations is therefore not a shortcoming of this instrument or a limit of this toy. It is what the quantities the audit is built from are able to express.

The same audit reads the declared covert channel at exactly 1 bit and reads a constant at exactly 0, so a modification that changes an output distribution is visible to it and a modification that changes only the cost of a computation is not. This is a statement about what quantities of this kind can express and it is not advice about inspecting anything. No real curve, standard, protocol, implementation, or deployed system is modeled or evaluated in any of the three registrations, the toy curve was chosen only because it can be enumerated exactly, and no attack was developed.

VII. ONLY A PROCEDURE'S DECLARED FREEDOM IS AUDITABLE BECAUSE THE ARTIFACT CARRIES NO RECORD OF THE SEARCH

Equation (6) says an audit built from informations cannot see a constant. What such an audit can see is provenance, and provenance turns out to be a counting question with three numbers in it, the rarity of a declared class, the number of distinct results a procedure can reach, and the product of the two.

The substrate is every non-singular curve over the field of 101 elements, 10100 of them, with the size of each group computed exactly by counting points. The declared class

is the set of curves whose group order has no prime factor above the declared bound of five. The registration names that class weak, and the name is a declaration rather than anything this run measures. What the run measures about the class is how rare it is and how many of its members each declared procedure can reach. It contains 2275 of the 10100 curves, a density of 0.2252.

Two declared procedures share one declared seed map, which sends a seed to a coefficient pair and rejects the singular pairs. The rigid procedure may use exactly one seed. It reaches 1 curve, its number of reachable results times the measured density is 0.2252, and it did not land in the declared class. The flexible procedure may use any seed in a declared range of 4096. Its declared range is not its freedom. Those 4096 seeds reach only 99 distinct curves, so the nominal freedom of the procedure overstates the freedom it actually has by a factor of more than forty, and the number that matters is the smaller one. Even so, 99 reachable results times a density of 0.2252 is 22.3, and a product above one is the point at which choosing among results is expected to reach the class. It does. The flexible procedure reaches the declared class at its tenth seed, and 526 of its 4096 seeds land there, an expected 7.787 seeds per success.

The published artifact carries none of that. It is a curve and a seed, and a seed reached on the first try and a seed reached on the thousandth try are the same kind of object with the same appearance, so the artifact records no search count and nothing in it distinguishes one history from another. What remains auditable is the freedom the procedure declared for itself. The measurement is about a declared toy parameter space with a declared toy class. It is not a finding about any real curve, standard, or generation procedure, none is modeled here, and nothing in it says that any procedure or parameter set was manipulated.

The connection was recorded in the track document with the result. Removing a designer's freedom to try again until the answer is the wanted one is what rigid parameter generation does in cryptography and what preregistration does in this campaign. They are one idea in two fields and this run computes the quantity that both of them bound. The campaign has recorded four failures of its own against exactly this, registrations whose declared bars could be met without the event they were meant to detect ever occurring, and the standing rule that came out of them requires every registration to cite a probe verifying that the event is present before a bar can mean anything. A bar chosen after the answer is known and a curve chosen after its properties are known are the same freedom being spent in two disciplines.

VIII. READING AN ENSEMBLE THAT IS NOT ORTHOGONAL COSTS TWO BITS OF INFORMATION FOR EACH UNIT OF INDUCED ERROR

The two sections above found a ceiling and a counting argument. This section and the next find the part of the arc where neither applies, because the security structure in question lives in distributions rather than in the cost of a computation, which is exactly where the campaign's instruments have no ceiling.

The campaign's hydrodynamic track measured what a restricted consumer loses when the substrate is exactly reversible. A coarse observer's mode decayed to a small fraction of its initial amplitude and the exact reverse of the dynamics returned the initial state bit for bit, so the substrate had forgotten nothing and only the observer had [15]. This section measures a declared model where that is not available, and it measures the price exactly.

The substrate is four declared preparations of one two-level system, two of them in one basis and two in the conjugate basis, chosen uniformly, so no measurement distinguishes all four. A declared consumer intercepts each preparation with probability λ , measures in the computational basis, and resends the state corresponding to its outcome, and the declared ladder for λ is 0, 0.1, 0.25, 0.5, 0.75, and 1. The receiving party measures in the preparing party's basis, and the induced error is the rate at which the receiving party's outcome differs from the prepared bit on the occasions when both used the same basis. Everything is enumerated exactly over the finite probability space. The declared preparation ensemble is the one Bennett and Brassard wrote down [9], and the reason a consumer cannot copy what it reads and pass the original along untouched is the no-cloning theorem [10].

All five declared items passed and Fig. 4 shows the run. The classical control is exact. On an orthogonal ensemble the same interception structure gives the consumer 1 full bit at an induced error of exactly 0, so reading without leaving a trace is available precisely when the states are distinguishable. On the declared ensemble every positive interception rate produces both positive information and positive error, and at rate zero both are exactly zero. The measured pairs match their declared closed forms, half the interception rate in bits and a quarter of it in error, with a worst deviation of 1.1×10^{-16} across the ladder, running from 0.05 bits at an error of 0.025 to 0.5 bits at an error of 0.25. The Holevo quantity of the declared preparation ensemble is 1 bit and bounds every measured value, the largest of which is 0.5 [11].

The finding declared with the bars is the exchange rate. Every rung of the ladder gives exactly 2 bits of information for each unit of induced error, to within 1.3×10^{-15} . The rate is a property of the declared preparation ensemble and the declared interception family and it is not a general constant, which is why the run declares both before measuring either.

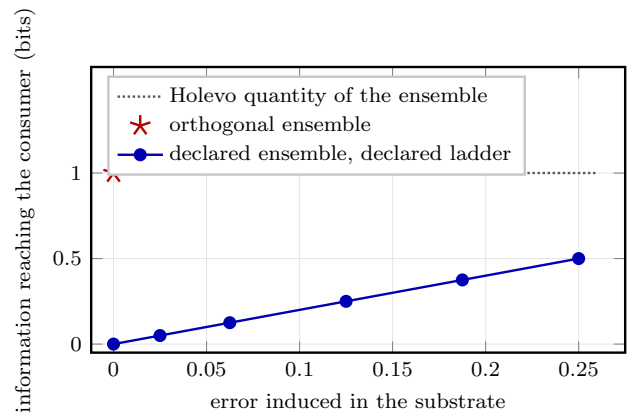


FIG. 4. Information against disturbance on the declared ladder. Each circle is one interception rate, plotted at the error the consumer induces in the substrate and the information it obtains. The six points lie on a straight line through the origin of slope exactly two bits per unit of induced error. The star on the vertical axis is the declared classical control, an orthogonal ensemble read perfectly at an induced error of exactly zero. The dotted line is the Holevo quantity of the declared preparation ensemble, 1 bit, which bounds every measured value. All values are read from the committed evidence record.

This is the first place in the program where a read is paid for by the substrate rather than by the reader. Where the substrate is exactly reversible a consumer that cannot see something loses the information and the substrate keeps it. Here a consumer that sees something has changed what it saw, by an amount fixed to the last digit by how much it saw. The declared model is a toy ensemble of four states. No protocol, device, implementation, or deployed system is modeled or evaluated, and no attack is developed.

IX. ONE CLASSICAL RECORD IS HELD IN FULL BY BOTH CONSUMERS AND ONE SHARED CORRELATION IS HELD BY ONE

The campaign's quantum track measured redundancy, meaning many fragments of an environment carrying the same record about a system, and it measured when redundancy appears. On the declared ladder of record strengths the redundancy of the record was 0 at every rung where the interaction wrote a weak record, with the fragment informations reading 0.0391, 0.1151, 0.2903, and 0.6122 bits, and it became 6 only at the rung where the interaction wrote a full record and the fragment information reached 1 bit. A declared control state with no preferred basis carried redundancy 0 at a system entropy of 0.9945 bits, so redundancy is a property of the declared interaction rather than of the entanglement. A third run measured two disjoint consumers agreeing about the system at rates 0.5096, 0.5373, 0.6357, 0.8750, and 1.0 along the same ladder, so agreement between observers is record strength

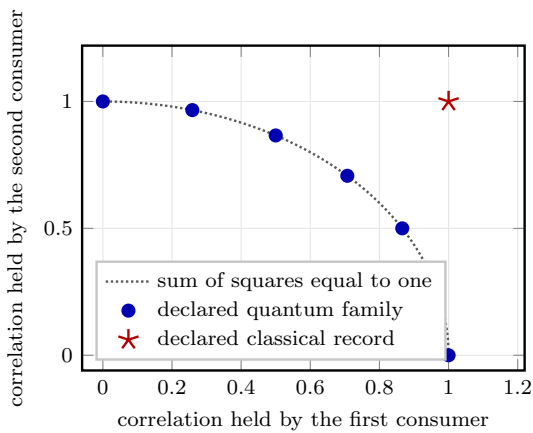


FIG. 5. What two consumers can hold at once. Each circle is one point of the declared angle grid, plotted by the correlation each consumer holds with the first party. The seven points lie on the quarter circle where the two squared correlations sum to one, so every unit one consumer gains is a unit the other loses, and at the two ends one consumer holds all of it and the other holds none. The star is the declared classical control, one bit of record held in full by both consumers at once, which sits outside the curve because a record can be copied. The two axes carry different quantities, a concurrence for the quantum family and a mutual information in bits for the classical record, and each is plotted against its own maximum of one. All values are read from the committed evidence record.

made mechanical [15]. This section measures the boundary of that plateau, meaning the information for which redundancy is not merely absent but unavailable.

Two declared substrates are compared. The classical one is a single bit written into a record and given to both consumers. The quantum one is a declared three-qubit family carrying one excitation shared between the two consumers at a fixed weight split, parametrized by an angle on a declared seven-point grid. Correlation between two parties is the Wootters concurrence of the exact reduced state [12], and the correlation of one party against the other two is its tangle.

All four declared items passed and Fig. 5 shows the run. The broadcast classical record gives each consumer the whole bit, both mutual informations exactly 1 and their sum exactly 2. The declared quantum family obeys its tradeoff as an identity rather than as an inequality. The sum of the two squared concurrences is 1 across the whole grid with a worst deviation of 1.9×10^{-10} , so the relation of Coffman, Kundu, and Wootters, which in general bounds the correlation two parties can share against a third, is saturated on this family to the same precision [13]. At one end of the grid the first consumer holds concurrence exactly 1 and the second holds exactly 0, and at the other end the second holds exactly 1 while the first measures 6.1×10^{-17} . The grid interpolates smoothly between the two ends and the equal split puts both at 0.7071. The deviations of 10^{-10} are the accumulated arithmetic of eigenvalues taken in double precision and they are the

checks the run declares, not a spread in the underlying quantities.

The classical informations sum to 2 while the quantum squared correlations sum to 1, and that is the whole comparison. Two consumers can hold one classical record in full and cannot hold the declared correlation at once. Redundancy, which is what makes a fact objective by letting many consumers read it independently and agree, is available exactly for information that has already become classical. The quantum track found the plateau and this run measures its boundary. The identity measured here is a property of the declared family on the declared grid and it is not a general identity, the general relation being an inequality. No protocol, device, implementation, or deployed system is modeled or evaluated, and the quantum track's declarations were fixed and run before this one was written.

X. WHAT THIS PAPER ESTABLISHES

Tables III and IV register the eleven runs with their recorded verdicts. Read together they answer the arc's question inside its declared scope, which of these models' security statements are bookkeeping relative to a declared consumer and which no declared consumer can move.

Three findings of the first movement are consumer-relative in the precise sense that a declared change moves them with the substrate held fixed. Which of two leakage channels is better depends on the reader's task whenever the two are incomparable, and incomparability is not rare, since 856 of 2000 declared random pairs were certified incomparable and 268 of them were ordered unanimously by every scalar measure in the battery while a declared task reversed that order. Which entropy prices a source depends on the consumer's task, and the same declared source costs 3.477 bits to a consumer who averages and exactly 1 bit to a consumer who guesses once. How a redundancy plateau approaches its height depends on whether a designer placed it, since the engineered plateau is a step with nothing in between while the plateau the quantum track found takes a different value at every fragment size, its height and its approach set by how strong the records are.

Two findings of the first movement are the opposite, properties no declared consumer moves. Below the declared threshold the secret-sharing scheme gives every reader exactly the prior, and that zero is exact rather than small, so no choice of task, budget, or reading strategy reaches the secret. And a garbling never helps any task, which held across every declared control here at the level of 10^{-16} , so wherever comparability is certified a scalar ordering is a task ordering. The constructive consequence is stated with the negative one and measured by the same runs. For two channels that read the same hidden quantity the certificate of Eq. (3) decides comparability by proof with no threshold, and an evaluation that establishes comparability may rely on its scalar ordering.

TABLE III. The ledger of the arc’s first movement, what an audit built from informations certifies. Verdicts are recorded exactly as computed from the declared items of each run, including the failure. Measured in model, exploratory label, means an exact measurement in a declared finite construction with append-only evidence records and bars, seeds, ladders, and batteries fixed before the run. Key numbers are quoted from the committed records. Table IV carries the other two movements and the scope of the whole arc.

Run	Verdict	Key measurements	Label
CR-0 secret-sharing plateau	PASS	information curve exactly $(0, 0, \log_2 31, \log_2 31, \log_2 31)$ bits over 29791 enumerated outcomes, all 15 below-threshold subsets exactly 0 with posterior deviation exactly 0, all 10 threshold subsets exactly $\log_2 31$ with conditional entropy exactly 0 and Lagrange recovery in 297910 of 297910 checks, maximum distance to the nearer plateau exactly 0, short-randomness control at $\log_2(31/5) = 2.6323$ bits per single share against a 0.1 bit bar, redundant-share control exactly 0 to every single reader	instrument
CR-1 leakage ordering	PASS	channel informations 4, 2.0306, 1, 0.8867, and 0 bits, every non-identity channel a garbling of the identity with conditional information exactly 0, declared noise a garbling at residual 2.2×10^{-16} with the reverse proved impossible at 1.1439 bits, Hamming weight and low bit proved incomparable at 1.8113 and 0.7806 bits, data-processing control over 201 tasks with worst gain 5.6×10^{-17}	instrument, and measured in model, exploratory
CR-2 scalar orderings against tasks	PASS	exhibit proved incomparable, conic bound 0.00175 and total-variation difference 0.095, all seven divergences ordering A above B by at least 0.0149 while a declared task prefers B by 0.00212, garbling control over 500 pairs with worst task gain 3.3×10^{-16} , ensemble of 2000 pairs giving 867 comparable, 856 incomparable, 277 ambiguous, 268 unanimous among the incomparable and 268 of those 268 carrying task reversals, zero reversals among comparable pairs	measured in model, exploratory
CR-3 two entropy accountings	FAIL	closed forms to 8.9×10^{-16} , single-guess identity to 1.4×10^{-17} , Kraft equality exact and Huffman margins 0.0 and 0.0807 bits, ordering and equality clauses held exactly, but the declared clause requiring the difference of the two accountings to increase along the ladder measured 0, 0.398, 1.337, 2.275, 2.635, 2.113, 0.599 bits and turned over, because both accountings vanish together as the source becomes certain	measured in model, exploratory
CR-3b corrected declaration	PASS	the unchanged items repeated their values, the ratio falls strictly along both ladders, 1.0 to 0.1464 and 0.2876 to 0.1112, the closed-form derivative changes sign exactly once inside the declared bracket and agrees with a central finite difference to 1.6×10^{-8} , bisection in 39 steps places the interior maximum at spike weight 0.35338 with separation 2.6398 bits	measured in model, exploratory

The second movement locates where these instruments stop, and the stop is a bound rather than a difficulty. A constant of a parameter set has zero entropy, so every quantity built from informations reads exactly zero about it, while the same quantities read a declared one-bit covert channel at exactly one bit. Whatever content a constant of that kind has is the cost of a computation, and this campaign measures no such thing anywhere. Where information-theoretic quantities do keep working is provenance, and the measurement is a count. A procedure’s declared freedom times a declared class’s measured rarity decides whether choosing among results reaches the class, it stood at 0.2252 for the rigid procedure and 22.3 for the flexible one, and the artifact each procedure publishes records nothing about how many results were tried. The declared range of the flexible procedure was 4096 and its reachable set was 99, so nominal freedom and effective freedom are different numbers.

The third movement is the part of the arc where none of that stopping occurs. A consumer reading a declared ensemble whose states are not all distinguishable obtains information only by changing what it read, at exactly two bits per unit of induced error on the declared family,

while the same structure on a distinguishable ensemble reads the whole bit and changes nothing. The campaign’s hydrodynamic track measured the opposite, an exactly reversible substrate returning a coarse observer’s decayed mode bit for bit. And a classical record is held in full by two consumers at once, informations exactly one and one, while a shared quantum correlation on the declared family obeys an exact identity that leaves a second consumer with nothing when the first holds all of it. Redundancy is available exactly for information that has already become classical.

Three registrations failed and all three are kept. A declared clause demanded that the number of bits separating two accountings increase along a ladder of sources, and the difference turned over because both accountings vanish together on a source that is almost certain, so the design error was in the choice of quantity rather than in the substrate or the instrument. A second registration asked whether an audit can separate two parameter sets that are not two classes of object, on a group too small for its own uniformity bar and with a recovery loop that accepted a candidate it had not finished testing. A third failed on a coding error in an inverse map, which

TABLE IV. The ledger of the arc’s second and third movements, where an audit built from informations stops and where it does not, with the scope of the whole arc in the last row. Verdicts are recorded exactly as computed from the declared items of each run, including the two failures. Key numbers are quoted from the committed records. The last row is a non-claim and governs every row above it and every row of Table III.

Run	Verdict	Key measurements	Label
CR-4 backdoor audit ceiling	FAIL	a group of 96 elements and 8 possible output words put the output entropies at 2.7890 and 2.8002 bits against a uniform 3, outside the declared 0.15 bit window by finite size alone, and the reader holding the scalar predicted at 0.263 against a declared 0.5 because the recovery loop accepted the first candidate consistent with one observed word, and the scalar of the declared-rule point was itself recovered by enumeration at 41, so the declared finding that the audit cannot separate the two parameter sets read False and was the wrong question	measured in model, exploratory
CR-4b corrected question	FAIL	on a group of prime order 1013 the positive control is exact, the declared one-bit covert channel carrying exactly 1 bit about the seed bit and the unmodified generator exactly 0, output entropies 2.9878 and 2.9943 bits against a uniform 3, but the two routes to one conditional entropy read 2.8811 and 2.9125 bits, a difference of 0.0314 against a 10^{-9} bar, from an inverse map that multiplied an abscissa as though it were a scalar	measured in model, exploratory
CR-4c corrected inverse map and exact bound	PASS	same substrate and same positive control, the two routes now agree at exactly 0, and the ceiling is recorded as a bound rather than as an agreement within tolerance, a trapdoor scalar being a constant of the parameter set with entropy exactly 0 and therefore an information bound of exactly 0 about any observable	measured in model, exploratory
CR-5 designer freedom against class rarity	PASS	exact over all 10100 non-singular curves, 2275 in the declared class for a density of 0.2252, the rigid procedure reaching 1 curve at a freedom times density of 0.2252 and not landing in the class, the flexible procedure’s declared range of 4096 seeds reaching only 99 distinct curves at a freedom times density of 22.3 and landing in the class at seed 10 with 526 of 4096 seeds landing there	measured in model, exploratory
CR-6 information against disturbance	PASS	the orthogonal control read at exactly 1 bit with induced error exactly 0, every positive interception rate giving both positive information and positive error and rate zero giving exactly zero of both, measured pairs matching their declared closed forms to 1.1×10^{-16} from 0.05 bits at error 0.025 up to 0.5 bits at error 0.25, the Holevo quantity at 1 bit bounding every value, and the exchange rate exactly 2 bits per unit of induced error at every rung	measured in model, exploratory
CR-7 one record for both consumers and one correlation for one	PASS	the broadcast classical record giving both consumers exactly 1 bit for a sum of exactly 2, the declared family’s squared concurrences summing to 1 across the whole grid with worst deviation 1.9×10^{-10} and the Coffman-Kundu-Wootters relation saturated to the same precision, and at the ends of the grid one consumer at concurrence exactly 1 leaving the other at 0 and at 6.1×10^{-17}	measured in model, exploratory
Scope	non-claim	the models here are exact finite toy models, this paper says nothing about computational hardness or asymptotic security, no real cipher, protocol, curve, standard, implementation, device, or deployed system is modeled or evaluated, no attack is developed, and every result is about evaluation methodology in declared finite models	non-claim

the record names, and measured that error rather than anything about its subject. Each correction is a separate registration with its own declared items, and the sequence of verdicts is part of the evidence rather than a step on the way to it.

These are structural results about declared finite models carrying the campaign’s exploratory label. They are about evaluation methodology. They do not show that any real cipher, protocol, curve, standard, implementation, device, or deployed system is strong or weak, no experiment reported here can bear on that question, and no attack was developed or evaluated. Two objects here do not depend on the substrates measured. The first is the certificate, which settles by proof whether a scalar

comparison between two declared finite channels that read the same hidden quantity is a comparison for every task. The second is Eq. (6), which is not a measurement at all but a bound on what the measurements of this kind can express.

XI. METHODS AND REPRODUCIBILITY

All runs follow the campaign’s standing discipline. Substrates, channels, tasks, ladders, batteries, seeds, tolerances, and falsification bars were declared in the track document before the runs they govern, per the anti-circularity contract, and verdicts are computed from the declared

items rather than written first. The one bar that moved before its run moved for the reason given in Sec. I, and the record carries both the moved bar and the abandoned object's measured values.

Every result-bearing run writes an append-only JSON evidence record containing the declared parameters, the measured outcomes, the computed verdict, and a canonical SHA-256 hash of the record body. The records for this paper are `cr0-instrument.json`, `cr1-leakage-ordering.json`, `cr2-countermeasure-flip.json`, `cr3-task-entropy.json`, `cr3b-task-entropy.json`, `cr4-backdoor-ceiling.json`, `cr4b-backdoor-ceiling.json`, `cr4c-backdoor-ceiling.json`, `cr5-rigidity.json`, `cr6-information-disturbance.json`, and `cr7-monogamy.json` in the campaign repository. Each has its committed runner beside it, named by replacing the hyphens of the record name with underscores and the suffix with `.py`. The three failed records are retained unchanged and each corrected declaration is a separate record, so the sequence of verdicts is itself part of the evidence. A reducer may summarize records but may not rewrite them.

Every measured number and every figure coordinate in this paper is read from those committed records. The figure tables are generated deterministically by a committed script, `make_figdata_cr.py`, which uses only the Python standard library and has no free parameters, and which asserts each value typed in this paper against the record value it came from before it writes anything, so a record that changed would break the build rather than disagree quietly with the text. The comparison curve of Fig. 1 is carried in the CR-0 record itself and is checked against

the quantum track's own record by the same script, and the quantum track's redundancy ladder and agreement curve quoted in Sec. IX and the hydrodynamic track's exact reversal quoted in Sec. VIII are checked against their own records the same way. The declared channel matrices, ladders, grids, rules, and costs quoted in the text are checked against the track document that declared them.

The CR-0 substrate was enumerated exactly with integer counting over all 29791 outcomes and the CR-3 and CR-3b substrates were built from exact rationals, so in those runs the only floating-point operations are the logarithms and the compensated sums that turn exact counts and exact rationals into bits. The CR-1 and CR-2 runs compute exact finite sums over declared joint distributions. The CR-2 comparability classification is made by the projected-gradient garbling search of the campaign's decision-theoretic instrument against the declared tolerances, and CR-1 uses that search only as a cross-check against the certificate. The three CR-4 runs and CR-5 do integer arithmetic on enumerated group elements and enumerated coefficient pairs, with every point addition and every group order computed by counting rather than by any closed form, and the curve orders of CR-5 are obtained by scanning the whole field for each of the 10100 curves. CR-6 and CR-7 enumerate finite state vectors in double precision and every measured quantity in them is checked against a declared closed form or a declared identity, so the deviations of 1.1×10^{-16} and 1.9×10^{-10} recorded in those runs are the values of those checks. All eleven runs executed on a single workstation and require no special hardware.

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